

Project Apex Race Report - Race Event

Executive Summary

The race was largely defined by a clear performance hierarchy, with the Porsche 992 platforms demonstrating superior raw pace and efficient tire management, while the CAY M and CAY P/A models struggled to match their performance. Leaders: The Porsche 992 P/A and 992 M classes were the dominant force. Car #295 (Loftus Motorsports, 992 P/A) set the fastest overall lap at 1:27.896 and maintained the best average green flag pace of 1:28.616. This class also exhibited excellent driver parity, with a minimal 0.394-second gap between fastest and slowest drivers within the manufacturer. Key to their success was also superior tire management, exemplified by Car #243's low degradation coefficient of 0.004094. The 992 M cars, led by Car #269 (Competition Motorsports) with a 1:27.946 fastest lap, closely followed, showcasing strong overall performance and consistency. Mid-field: This tier comprised some of the less optimized 992 M cars and the stronger CAY M and CAY P/A entries. Car #204 (Paradaime Engineering, 992 M), despite being a 992, showed a notable weakness in tire degradation (0.023964 `deg_coeff_a`) and a slower average pace of 1:29.566, hindering its potential. Among the CAY models, Car #49 (Tatum Racing, CAY M) demonstrated the best pace for its class at 1:33.259 and commendable driver consistency (0.547 STDEV), indicating that while the platform lacks outright speed, individual driver skill and setup can yield competitive results within its tier. Car #78 (HM Road Racing, CAY P/A) also showed promising tire degradation (0.004831 `deg_coeff_a`) despite its overall pace deficit. Laggards: The tail end of the field was occupied by several CAY M cars, notably Car #43 (CHR) and Car #44 (Tatum Racing). Car #43 recorded the slowest average green flag pace at 1:37.293 and struggled with consistency (2.428 STDEV). Car #44 was particularly problematic, exhibiting an alarming race pace consistency standard deviation of 9.361, pointing to significant underlying issues, likely driver-related or a fundamental setup flaw. The CAY M manufacturer overall suffered from the largest driver pace gap of 2.420 seconds, highlighting a lack of consistent performance across its entries. The biggest strategic differentiator in this race was the synergy between raw vehicle performance and optimized tire management, allowing the leading Porsche 992 teams to consistently extract maximum performance throughout their stints.

Tactical Insights

- `{'tier': 'Leading Team', 'recommendation': "For Loftus Motorsports (Car #295, Porsche 992 P/A), while leading in overall pace (1:28.616 average green pace) and setting the fastest lap`

- (1:27.896), marginal gains can still be found in tire degradation. Car #243 (Competition Motorsports, 992 P/A) demonstrated a slightly superior tire degradation coefficient (0.004094 vs. #295's 0.006396). Investigate #243's setup and driving style, particularly in the latter stages of stints, to further minimize tire wear and potentially extend stint length or maintain peak performance for longer, converting current good performance into an even more dominant advantage.", 'justification': "Car #295's average green pace is 1:28.616 and fastest lap is 1:27.896. Car #295's tire degradation coefficient (deg_coeff_a) is 0.006396, while Car #243's is 0.004094, indicating better tire management potential."}
- {'tier': 'Mid-field Team', 'recommendation': "Paradaime Engineering (Car #204, Porsche 992 M) is hampered by the highest tire degradation among competitive 992 cars, with a deg_coeff_a of 0.023964, significantly higher than the class average. This leads to a predicted final 5-lap loss of 5.401 seconds. Focus engineering efforts on understanding and mitigating this excessive tire wear. Analyze driver Mark Crigler's inputs and car setup for potential over-driving or sub-optimal mechanical balance that is prematurely degrading the tires. Reducing this degradation is critical to improving their 1:29.566 average green pace and moving up the field.", 'justification': 'Car #204 exhibits a high tire degradation coefficient (deg_coeff_a: 0.023964) with a predicted final 5-lap loss of 5.401 seconds, impacting its 1:29.566 average green pace.'}
 - {'tier': 'Lagging Team', 'recommendation': "For CHR (Car #43, Porsche CAY M), the primary focus must be on improving fundamental driver consistency and overall pace. Paul Bonderson's average green pace of 1:37.293 is the slowest in the field, coupled with a high race pace consistency STDEV of 2.428. While some CAY M cars like #49 (Greg Herback) show better consistency (0.547 STDEV) and pace (1:33.259), the significant 2.420-second driver pace gap within the CAY M manufacturer indicates a widespread issue. Implement a targeted driver coaching program for Paul Bonderson focusing on consistent lap times and traffic management, and conduct a thorough car setup review to ensure the vehicle is stable and predictable, allowing for more consistent driver performance.", 'justification': "Car #43's average green pace is 1:37.293 and its race pace consistency STDEV is 2.428. The CAY M manufacturer has the largest driver pace gap of 2.420 seconds. In comparison, Car #49 (CAY M) has an average green pace of 1:33.259 and a consistency STDEV of 0.547."}